

Algorithm for file updates in Python

Project description

Develop an algorithm that parses a file containing IP addresses that are allowed to access restricted content and removes addresses that no longer have access.

Open the file that contains the allow list

```
In [2]: # Assign `import_file` to the name of the file
import_file = "allow_list.txt"

# Assign `remove_list` to a list of IP addresses that are no longer allowed to access restricted information.
remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# First line of `with` statement
with open(import_file, "r") as file:
```

`with` handles errors & manages external resources for us.

`open()` used to open the file that we needed, and the second parameter is either “w”, “r”, “a” depending on the mode that we want to open the file (write, read, append)

Read the file contents

```
In [3]: # Assign 'import_file' to the name of the file
import_file = "allow_list.txt"

# Assign 'remove_list' to a list of IP addresses that are no longer allowed to access restricted information.
remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build 'with' statement to read in the initial contents of the file
with open(import_file, "r") as file:
    # Use '.read()' to read the imported file and store it in a variable named 'ip_addresses'
    ip_addresses = file.read()

# Display 'ip_addresses'
print(ip_addresses)

ip_address
192.168.25.60
192.168.205.12
192.168.97.225
192.168.6.9
192.168.52.90
192.168.158.170
192.168.90.124
192.168.186.176
192.168.133.188
192.168.203.198
192.168.201.40
192.168.218.219
192.168.52.37
192.168.156.224
192.168.60.153
192.168.58.57
192.168.69.116
```

`.read()` states that the file variable will be opened in read mode

Convert the string into a list

```
In [4]: # Assign 'import_file' to the name of the file
import_file = "allow_list.txt"

# Assign 'remove_list' to a list of IP addresses that are no longer allowed to access restricted information.
remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build 'with' statement to read in the initial contents of the file
with open(import_file, "r") as file:
    # Use '.read()' to read the imported file and store it in a variable named 'ip_addresses'
    ip_addresses = file.read()

# Use '.split()' to convert 'ip_addresses' from a string to a list
ip_addresses = ip_addresses.split()

# Display 'ip_addresses'
print(ip_addresses)

['ip_address', '192.168.25.60', '192.168.205.12', '192.168.97.225', '192.168.6.9', '192.168.52.90', '192.168.158.170', '192.168.90.124', '192.168.186.176', '192.168.133.188', '192.168.203.198', '192.168.201.40', '192.168.218.219', '192.168.52.37', '192.168.156.224', '192.168.60.153', '192.168.58.57', '192.168.69.116']
```

Iterate through the remove list

```
In [6]: # Assign 'import_file' to the name of the file
import_file = "allow_list.txt"

# Assign 'remove_list' to a list of IP addresses that are no longer allowed to access restricted information.
remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build 'with' statement to read in the initial contents of the file
with open(import_file, "r") as file:

    # Use '.read()' to read the imported file and store it in a variable named 'ip_addresses'
    ip_addresses = file.read()

# Use '.split()' to convert 'ip_addresses' from a string to a list
ip_addresses = ip_addresses.split()

# Build iterative statement
# Name loop variable 'element'
# Loop through 'ip_addresses'
for element in ip_addresses:

    # Display 'element' in every iteration
    print(element)

ip_address
192.168.25.60
192.168.205.12
192.168.97.225
192.168.6.9
192.168.52.90
192.168.158.170
192.168.90.124
192.168.186.176
192.168.133.188
192.168.203.198
192.168.201.40
192.168.218.219
192.168.52.37
192.168.156.224
192.168.60.153
192.168.58.57
192.168.69.116
```

Remove IP addresses that are on the remove list

```
In [9]: # Assign 'import_file' to the name of the file
import_file = "allow_list.txt"

# Assign 'remove_list' to a list of IP addresses that are no longer allowed to access restricted information.
remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build 'with' statement to read in the initial contents of the file
with open(import_file, "r") as file:

    # Use '.read()' to read the imported file and store it in a variable named 'ip_addresses'
    ip_addresses = file.read()

# Use '.split()' to convert 'ip_addresses' from a string to a list
ip_addresses = ip_addresses.split()

# Build iterative statement
# Name loop variable 'element'
# Loop through 'ip_addresses'
for element in ip_addresses:

    # Build conditional statement
    # If current element is in 'remove_list',
    if element in remove_list:

        # then current element should be removed from 'ip_addresses'
        ip_addresses.remove(element)

# Display 'ip_addresses'
print(ip_addresses)

['ip_address', '192.168.25.60', '192.168.205.12', '192.168.6.9', '192.168.52.90', '192.168.90.124', '192.168.186.176', '192.168.133.188', '192.168.203.198', '192.168.218.219', '192.168.52.37', '192.168.156.224', '192.168.60.153', '192.168.69.116']
```

Update the file with the revised list of IP addresses

```
In [10]: # Assign 'import_file' to the name of the file
import_file = "allow_list.txt"

# Assign 'remove_list' to a list of IP addresses that are no longer allowed to access restricted information.
remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build 'with' statement to read in the initial contents of the file
with open(import_file, "r") as file:

    # Use '.read()' to read the imported file and store it in a variable named 'ip_addresses'
    ip_addresses = file.read()

# Use '.split()' to convert 'ip_addresses' from a string to a list
ip_addresses = ip_addresses.split()

# Build iterative statement
# Name loop variable 'element'
# Loop through 'ip_addresses'
for element in ip_addresses:

    # Build conditional statement
    # If current element is in 'remove_list',
    if element in remove_list:

        # then current element should be removed from 'ip_addresses'
        ip_addresses.remove(element)

# Convert 'ip_addresses' back to a string so that it can be written into the text file
ip_addresses = " ".join(ip_addresses)

# Build 'with' statement to rewrite the original file
with open(import_file, "w") as file:

    # Rewrite the file, replacing its contents with 'ip_addresses'
    file.write(ip_addresses)
```

`.write()` writes in the file the passed argument `ip_addresses`.

`.split()` used to change the string into a list

`.remove()` used to remove from a string

`.join()` used to convert the list into a string.

Lastly, the `for` loop, loops in `ip_addresses` checking whether `element` is in the remove list or not.

If element is in the `remove_list` we remove the ip from the `ip_addresses`.

```

In [11]: # Assign 'import_file' to the name of the file
import_file = "allow_list.txt"

# Assign 'remove_list' to a list of IP addresses that are no longer allowed to access restricted information.
remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build 'with' statement to read in the initial contents of the file
with open(import_file, "r") as file:

    # Use '.read()' to read the imported file and store it in a variable named 'ip_addresses'
    ip_addresses = file.read()

# Use '.split()' to convert 'ip_addresses' from a string to a list
ip_addresses = ip_addresses.split()

# Build iterative statement
# Name loop variable 'element'
# Loop through 'ip_addresses'
for element in ip_addresses:

    # Build conditional statement
    # If current element is in 'remove_list',
    if element in remove_list:

        # then current element should be removed from 'ip_addresses'
        ip_addresses.remove(element)

# Convert 'ip_addresses' back to a string so that it can be written into the text file
ip_addresses = " ".join(ip_addresses)

# Build 'with' statement to rewrite the original file
with open(import_file, "w") as file:

    # Rewrite the file, replacing its contents with 'ip_addresses'
    file.write(ip_addresses)

# Build 'with' statement to read in the updated file
with open(import_file, "r") as file:

    # Read in the updated file and store the contents in 'text'
    text = file.read()

# Display the contents of 'text'
print(text)

ip_address 192.168.25.60 192.168.205.12 192.168.6.9 192.168.52.90 192.168.90.124 192.168.186.176 192.168.133.188 19
2.168.203.198 192.168.218.219 192.168.52.37 192.168.156.224 192.168.60.153 192.168.69.116

```

In [15]: *# Define a function named 'update_file' that takes in two parameters: 'import_file' and 'remove_list'
and combines the steps you've written in this lab leading up to this*

```
def update_file(import_file, remove_list):  
  
    # Build 'with' statement to read in the initial contents of the file  
  
    with open(import_file, "r") as file:  
  
        # Use '.read()' to read the imported file and store it in a variable named 'ip_addresses'  
  
        ip_addresses = file.read()  
  
        # Use '.split()' to convert 'ip_addresses' from a string to a list  
  
        ip_addresses = ip_addresses.split()  
  
        # Build iterative statement  
        # Name loop variable 'element'  
        # Loop through 'ip_addresses'  
  
        for element in ip_addresses:  
  
            # Build conditional statement  
            # If current element is in 'remove_list',  
  
            if element in remove_list:  
  
                # then current element should be removed from 'ip_addresses'  
  
                ip_addresses.remove(element)  
  
        # Convert 'ip_addresses' back to a string so that it can be written into the text file  
  
        ip_addresses = " ".join(ip_addresses)  
  
        # Build 'with' statement to rewrite the original file  
  
        with open(import_file, "w") as file:  
  
            # Rewrite the file, replacing its contents with 'ip_addresses'  
  
            file.write(ip_addresses)  
  
        # Call 'update_file()' and pass in "allow_list.txt" and a list of IP addresses to be removed  
  
        update_file("allow_list.txt", remove_list)  
  
        # Build 'with' statement to read in the updated file  
  
        with open("allow_list.txt", "r") as file:  
  
            # Read in the updated file and store the contents in 'text'  
  
            text = file.read()  
  
            # Display the contents of 'text'  
  
            print(text)
```

```
ip_address 192.168.25.60 192.168.205.12 192.168.6.9 192.168.52.90 192.168.90.124 192.168.186.176 192.168.133.188 19  
2.168.203.198 192.168.218.219 192.168.52.37 192.168.156.224 192.168.60.153 192.168.69.116
```

Summary

Summarizing this lab incorporated everything I learned to build an algorithm to remove IP addresses from our allowed_list.